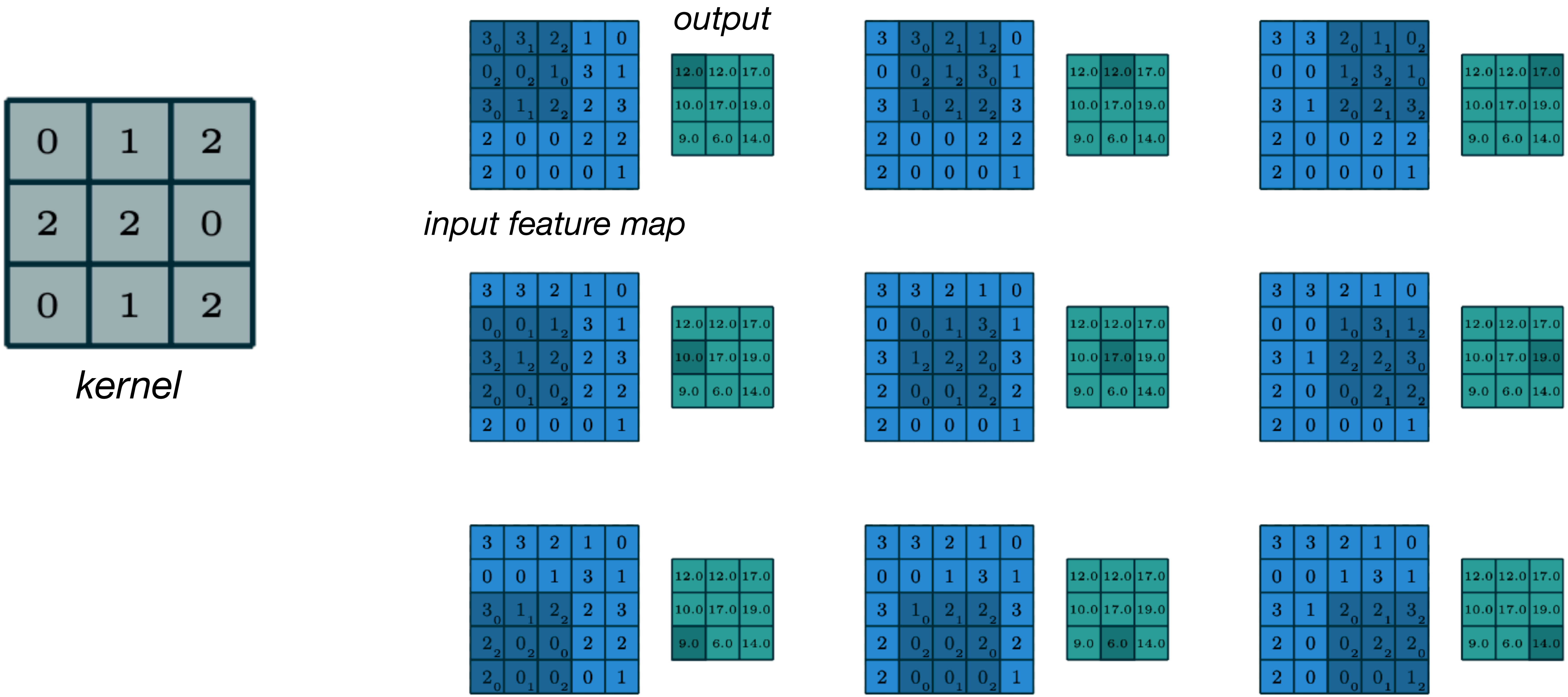
Discrete convolution



Dumouilin and Visin 2018

Figure 1.1: Computing the output values of a discrete convolution.

Discrete convolution

For input X and kernel K

$$X = \begin{bmatrix} x_{11} & x_{12} & x_{13} & x_{14} \\ x_{21} & x_{22} & x_{23} & x_{24} \\ x_{31} & x_{32} & x_{33} & x_{34} \\ x_{41} & x_{42} & x_{43} & x_{44} \end{bmatrix}, \qquad K = \begin{bmatrix} k_{11} & k_{12} & k_{13} \\ k_{21} & k_{22} & k_{23} \\ k_{31} & k_{32} & k_{33} \end{bmatrix}.$$

our output size with no padding and stride 1 will be:

$$(4 - 3 + 1) \times (4 - 3 + 1) = 2 \times 2$$

the general formula for the convolution is:

$$Y_{ij} = \sum_{m=1}^{3} \sum_{n=1}^{3} X_{i+m-1,\, j+n-1} \cdot K_{mn}.$$